

DIGITAL ATTENTION SPAN ANALYSER

A PROJECT REPORT

Submitted by

JAYASAKTHI R (8115U24AM027)

in partial fulfilment for the award of the course

AMB 1221-MACHINE LEARNING FUNDAMENTALS

IN

DEPARTMENT OF

COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)



**K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
SAMAYAPURAM, TRICHY**



**ANNA UNIVERSITY
CHENNAI 600 025**

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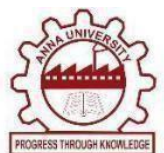
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BONAFIDE CERTIFICATE

Certified that this project report titled “**DIGITAL ATTENTION SPAN ANALYZER**” is the Bonafide work of **JAYASAKTHI R (8115U24AM027)** who carried out the work under my supervision.

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DECLARATION BY THE CANDIDATE

I declare that to the best of my knowledge the work reported here in has been composed solely by myself and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva-Voice held at K. Ramakrishnan College of Engineering on _____

SIGNATURE OF THE CANDIDATE

ACKNOWLEDGEMENT

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ANNA UNIVERSITY, CHENNAI



DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

VISION

To become a renowned hub for AIML technologies to producing highly talented globally recognizable technocrats to meet industrial needs and societal expectation.

MISSION

Mission of the Department

- M1** To impart advanced education in AI and Machine Learning, built upon a foundation in Computer Science and Engineering.
- M2** To foster Experiential learning equips students with engineering skills to tackle real-world problems.
- M3** To promote collaborative innovation in AI, machine learning, and related research and development with industries.
- M4** To provide an enjoyable environment for pursuing excellence while upholding strong personal and professional values and ethics.

PROGRAM EDUCATIONAL OBJECTIVES (PEO's)

- PEO1** Excel in technical abilities to build intelligent systems in the fields of AI & ML in order to find new opportunities.
- PEO2** Embrace new technology to solve real-world problems, whether alone or as a team, while prioritizing ethics and societal benefits.
- PEO3** Accept lifelong learning to expand future opportunities in research and product development.

PROGRAM SPECIFIC OUTCOMES (PSO's)

- PSO1** Expertise in tailoring ML algorithms and models to excel in designated applications and fields.
- PSO2** Ability to conduct research, contributing to machine learning advancements and innovations that tackle emerging societal challenges.

PROGRAM OUTCOMES (POs)

Engineering students will be able to:

PO1 Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2 Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3 Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4 Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5 Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6 The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7 Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO8 Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings

PO9 Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO10 Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO11 Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

The Digital Attention Span Analyzer using Artificial Intelligence is a smart system designed to monitor and analyze a user's focus level in real-time. In today's digital world, distractions such as social media, mobile usage, and overthinking significantly reduce productivity. This project aims to address this issue by developing an intelligent system that tracks user attention, analyzes behavior patterns, and provides feedback to improve focus. The system is implemented using Python and Streamlit, providing an interactive interface where users can log in, track their attention span, and analyze their daily routine. The system collects data such as productive time, distraction time, and idle time, and computes an attention score based on user behavior. Unlike traditional systems, this project integrates AI-based analysis (future scope: machine learning models) to predict user focus levels and provide personalized suggestions. Visualization tools such as charts and analytics dashboards are used to present insights clearly. This system helps users improve productivity, build discipline, and understand their focus patterns, making it highly useful for students and professionals.

ABSTRACT WITH POs AND PSOs MAPPING

Abstract	POs mapped	PSOs mapped
The Student Course Recommendation System is a Machine Learning-based application designed to assist students in selecting courses that best match their skills, interests, academic performance, and learning preferences. The system applies content-based filtering and similarity analysis to compare student profiles with course descriptions, generating personalized recommendations. A Graphical User Interface (GUI) built with Python provides an interactive platform where students can enter their details and instantly receive tailored course suggestions. This system promotes personalized learning, reduces course mismatch, and supports academic and career development.	PO1-3 PO2-3 PO3-3 PO4-3 PO5-3 PO6-3 PO7-3 PO8-3 PO9-3 PO10-3 PO11-3	PSO1-3 PSO2-3

Note: 1- Low, 2-Medium, 3- High

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LIST OF ABBREVIATIONS

S.NO	ACRONYM	ABBREVIATION
1	UI	User Interface
2	HCI	Human-Computer Interaction
3	EDA	Exploratory Data Analysis
4	GUI	Graphical User Interface
5	ML	Machine Learning

CHAPTER 1

INTRODUCTION

1.1 Objective

The main objective of the **Digital Attention Span Analyzer using Artificial Intelligence** is to develop a system that can monitor, evaluate, and improve the attention levels of users in a digital environment. In the present generation, excessive use of mobile phones, social media platforms, and multitasking has led to a decrease in concentration levels. This project aims to address this issue by providing a structured platform where users can track their focus and understand how their time is being utilized.

Another important objective of this system is to distinguish between productive time and distraction time during a session. By analyzing these parameters, the system calculates an attention score that reflects the user's level of focus. This score is further categorized into levels such as High, Medium, and Low, helping users easily interpret their performance. The system also provides suggestions based on routine analysis, enabling users to improve their habits and reduce distractions.

In addition, the project is designed with future enhancements in mind, particularly the integration of machine learning techniques. With real-time data collection, the system can be upgraded to predict user behavior and provide personalized recommendations. Thus, the objective of this project is not only to track attention but also to create an intelligent solution that enhances productivity and focus.

1.2 Overview

The Digital Attention Span Analyzer is a web-based application developed using Python and Streamlit, which provides an interactive and user-friendly platform for analyzing attention levels. The system consists of multiple modules such as Home, Routine Analysis, Motivation, Tracking, Analytics, and Resources, each designed to

perform a specific function. The application starts with a login interface, after which users can navigate through different features using a menu-driven interface.

The most important component of the system is the Attention Tracking module. In this module, users can start a timed session during which the system monitors their focus. Based on simulated logic, the system records productive and distraction time and calculates an attention score. The results are displayed using metrics and visual indicators, making it easy for users to understand their performance. The Analytics module further enhances this by presenting data in the form of charts, allowing users to track their progress over time.

Additionally, the system includes a Routine Analysis module that evaluates user habits such as sleep hours, work duration, and common distractions. The Motivation module provides encouraging messages to help users stay consistent and disciplined. Overall, the system acts as a complete platform for monitoring attention and improving productivity through interactive and data-driven features.

1.3 Purpose and Importance

The purpose of the Digital Attention Span Analyzer is to help individuals overcome the problem of reduced attention span in the digital age. With constant exposure to digital devices and online content, maintaining focus has become increasingly difficult. This project provides a solution by enabling users to monitor their attention levels and identify patterns of distraction, thereby promoting better concentration.

The importance of this project lies in its real-world application, especially for students and professionals. Students often face difficulty in concentrating on their studies due to various distractions, which affects their academic performance. By using this system, they can track their focus levels, identify weak areas, and take corrective measures. Similarly, professionals can use the system to improve their work efficiency and manage their time effectively.

Moreover, the project emphasizes the role of artificial intelligence in solving everyday problems. It demonstrates how user behavior can be analyzed using digital tools to provide meaningful insights. In the long run, such systems can be integrated into educational and professional environments to enhance productivity, making this project highly relevant and important.

1.4 Data Source Description

The data used in the Digital Attention Span Analyzer is primarily generated dynamically based on user interaction with the system. During each tracking session, the system records parameters such as productive time, distraction time, and session duration. This data is stored temporarily and used to calculate attention scores and generate visual representations.

In addition to tracking data, the system collects user inputs related to daily routines, including sleep hours, work duration, and types of distractions. This information is used in the Routine Analysis module to provide suggestions for improving focus and productivity. The collected data is simple but effective in understanding user behavior and identifying patterns.

For future enhancements, the system can incorporate real-world data sources such as screen time tracking, application usage logs, and user activity monitoring. This data can be stored in structured formats like CSV files or databases and processed using machine learning techniques. With proper data preprocessing and analysis, the system can evolve into a more advanced AI-based attention prediction tool.

1.5 Project Summarization

The Digital Attention Span Analyzer using Artificial Intelligence is a system developed to monitor and improve user attention in a digital environment. It

combines multiple modules such as tracking, routine analysis, motivation, and analytics to provide a complete solution for productivity improvement. The system calculates an attention score based on user activity and presents the results in an easy-to-understand format.

One of the key features of the system is its ability to visualize user performance using charts and graphs. This helps users track their progress over time and identify trends in their behavior. The user-friendly interface ensures smooth interaction, making the system accessible even to non-technical users. The inclusion of motivational messages further enhances user engagement and encourages consistent usage.

Although the current implementation uses simulated data, the project lays a strong foundation for future improvements. With the integration of machine learning, the system can provide predictive insights and personalized recommendations. Overall, the project successfully demonstrates how technology can be used to analyze attention and improve productivity, making it a valuable tool for students and professionals.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction and Traditional Methods

In the modern digital environment, maintaining attention and focus has become a major challenge due to the increasing use of smartphones, social media platforms, and online content. Individuals, especially students, are constantly exposed to distractions that reduce their productivity and efficiency. Traditionally, attention management was handled using manual techniques such as maintaining schedules, setting alarms, and following time management methods like the Pomodoro technique. These approaches relied heavily on self-discipline and lacked any form of automated monitoring.

Traditional methods of attention tracking did not involve any intelligent system to analyze user behavior. People depended on self-observation or external guidance from teachers and mentors to improve their focus. While these methods were simple, they were not always accurate or effective, as they could not measure real-time attention levels or provide data-driven insights. As a result, users often found it difficult to identify the exact causes of distraction.

With the advancement of technology, there is a growing need for automated systems that can monitor attention levels and provide meaningful feedback. This has led to the development of digital solutions that use data analysis and artificial intelligence to track user behavior and improve productivity. The proposed project builds upon this concept by introducing a system that analyzes attention in a structured and interactive manner.

2.2 Machine Learning Techniques for Recommendation Systems

Machine learning techniques play an important role in analyzing user behavior

and predicting attention patterns. Various algorithms such as classification, clustering, and regression can be used to identify whether a user is focused or distracted based on input data. These models learn from historical data and provide predictions that help in improving decision-making and performance.

In attention analysis systems, features such as time spent on tasks, frequency of interruptions, and user activity patterns are used as inputs for machine learning models. Algorithms like Decision Trees, Logistic Regression, and Support Vector Machines can classify user behavior into categories such as productive or distracted. More advanced techniques like Neural Networks and Deep Learning can also be used to detect complex patterns and provide highly accurate predictions.

Although the current project uses simulated data for attention tracking, it can be extended by integrating machine learning models in the future. With real-time data collection, the system can learn user behavior, predict attention levels, and provide personalized suggestions. This makes machine learning an essential component for enhancing the accuracy and intelligence of attention analysis systems.

2.3 Web-Based Systems and Visualization Techniques

Web-based systems have become increasingly popular due to their accessibility and ease of use. Applications developed using frameworks like Streamlit, Flask, or Django allow users to interact with systems through a simple interface without requiring complex installations. These systems provide real-time interaction and are suitable for applications such as attention tracking and productivity analysis.

Visualization techniques play a crucial role in presenting data in an understandable manner. Instead of displaying raw data, charts such as bar graphs, line graphs, and area charts are used to represent user performance visually. This helps users quickly identify patterns, trends, and areas that require improvement. Interactive dashboards further enhance user experience by allowing dynamic exploration of data.

In this project, Streamlit is used to develop a web-based interface that includes visual elements such as progress bars and charts. These visualizations make it easier

for users to interpret their attention scores and track their progress over time. Thus, web-based systems combined with effective visualization techniques provide a powerful platform for analyzing and improving attention.

2.4 Limitations of Existing Systems and Summary

Despite the availability of various attention tracking and productivity tools, many existing systems have certain limitations. Traditional methods lack automation and depend heavily on user discipline, while some digital tools provide only basic tracking without detailed analysis. Many systems also fail to provide personalized insights, making them less effective in improving user behavior.

Another major limitation is the lack of real-time data analysis and prediction. Some systems do not use machine learning techniques, which restricts their ability to understand complex user behavior patterns. Additionally, existing tools may not provide proper visualization or user-friendly interfaces, making it difficult for users to interpret the results.

The proposed Digital Attention Span Analyzer addresses these limitations by providing an interactive platform that combines tracking, analysis, and visualization. Although it currently uses simulated data, it is designed to support future integration of machine learning for improved accuracy and personalization. Overall, the project aims to overcome the drawbacks of existing systems and provide a more effective solution for attention analysis.

CHAPTER 3

PROJECT METHODOLOGY

The Project Methodology of the Digital Attention Span Analyzer explains the overall approach used to design and develop the system. It focuses on how different modules are structured and integrated to analyze user attention effectively. The system is designed using a simple and interactive approach so that users can easily monitor their focus and understand their productivity levels.

3.1 Proposed Workflow

The proposed workflow of the Digital Attention Span Analyzer begins with the user accessing the application and entering their name through the login interface. Once logged in, the user is directed to the main dashboard, where different modules such as Routine Analysis, Motivation, Tracking, and Analytics are available. Each module is designed to handle a specific aspect of attention analysis, allowing the user to interact with the system in a structured and organized manner.

The core workflow is centered around the Attention Tracking module. In this module, the user selects a duration and initiates the tracking process. During this session, the system continuously simulates user activity and records productive and distraction time. At the end of the session, an attention score is calculated using these values and displayed along with performance levels such as High, Medium, or Low. The collected data is stored and later used by the Analytics module to generate visual charts, helping users understand their performance trends and improve their focus over time. After completing the session, the generated data is stored in session history. This data is used by the Analytics module to display charts and help users understand their performance trends. The workflow ensures smooth execution from input to output, making the system easy to use and effective.

3.2 Architectural Overview

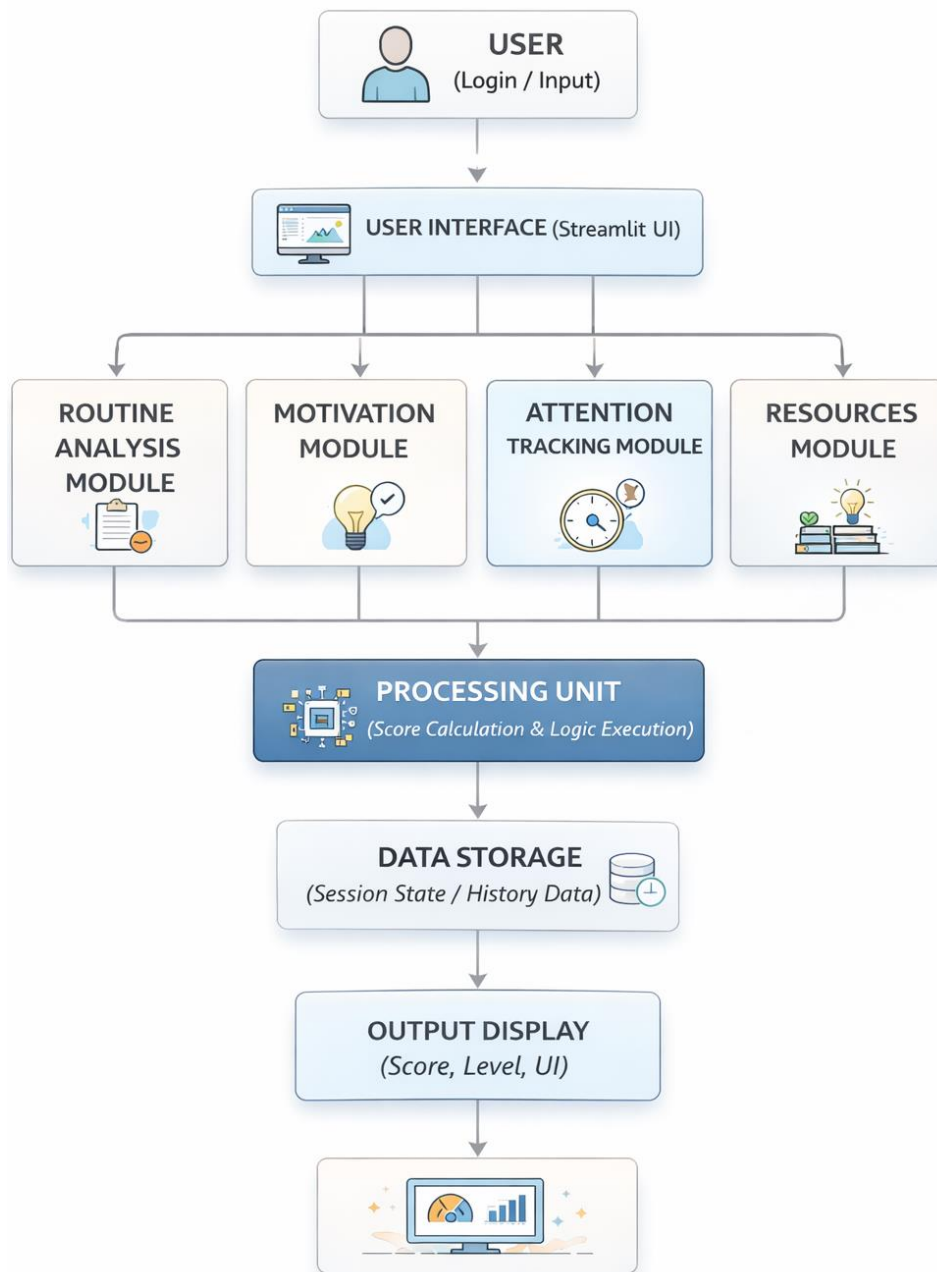


Fig 3.2.1 Architecture Diagram

The architecture of the Digital Attention Span Analyzer is designed in a modular and efficient manner, consisting of three main components: the User Interface, the Processing Unit, and the Data Handling module. These components are interconnected and work together to ensure smooth functioning of the system.

The modular design makes the system easy to understand, maintain, and extend with additional features in the future. The User Interface, developed using Streamlit, is responsible for collecting user inputs and displaying outputs such as attention scores, suggestions, and visualizations. The Processing Unit handles the core logic of the system, including attention tracking, score calculation, and routine analysis. The Data Handling module manages session data and stores user history for future reference. This architecture is lightweight and flexible, making it suitable for academic projects while also allowing future integration of databases and machine learning models.

3.3 Hardware and Software Requirements

The Digital Attention Span Analyzer requires minimal hardware resources, making it easy to implement and use. A standard computer or laptop with at least 4 GB RAM, a dual-core processor, and basic storage capacity is sufficient to run the application smoothly. Since the system is lightweight and does not involve heavy computations, it can operate efficiently without requiring advanced hardware configurations.

On the software side, the project is developed using Python as the primary programming language. Streamlit is used to create the interactive web interface, while pandas is used for data manipulation and analysis. Additional libraries such as time and random are used to simulate tracking behavior during attention sessions. The use of open-source tools ensures that the system is cost-effective, flexible, and easy to deploy across different platforms such as Windows, Linux, and macOS.

3.4 System Integration and Working

The system integrates multiple modules to provide a complete and efficient attention analysis solution. After logging into the application, the user can navigate through different modules that work together to analyze attention and provide feedback. Each module performs a specific function, and all modules are interconnected to ensure smooth data flow and consistent performance.

During the Attention Tracking process, the system monitors user activity by simulating productive and distraction time. Once the session is completed, an attention score is calculated and displayed along with performance levels. The generated data is stored in session history and used by the Analytics module to create visual representations such as charts. Additionally, the Routine Analysis module evaluates user inputs and provides suggestions for improvement. This integration of modules ensures accurate results, user-friendly interaction, and an effective approach to improving attention and productivity.

CHAPTER 4

RELEVANCE OF THE PROJECT

4.1 Why the Model Was Chosen

The Digital Attention Span Analyzer model was chosen to address the increasing problem of reduced focus in today's digital environment. With the rapid growth of smartphones, social media, and online platforms, individuals are constantly exposed to distractions that negatively impact their productivity. Traditional methods of improving focus, such as time management techniques, are not always effective because they lack real-time monitoring and analysis. Therefore, a digital system that can track and evaluate attention levels becomes essential.

The proposed model is designed using a simple yet effective approach that combines user interaction, simulated tracking, and data analysis. Instead of relying on complex hardware or expensive tools, the system uses software-based techniques to monitor attention. This makes the model easy to implement, cost-effective, and accessible to a wide range of users, especially students. The use of Streamlit further simplifies the development process by providing an interactive interface for users to engage with the system.

Another important reason for choosing this model is its scalability and flexibility. Although the current implementation uses simulated data, it can be extended to include real-time tracking and machine learning algorithms. This allows the system to evolve into a more advanced attention analysis tool in the future. Hence, the chosen model is not only practical for the present but also adaptable for future enhancements.

4.2 Comparison with Other Models

There are several existing models and tools designed to improve productivity and focus, such as time management applications and screen time trackers. However, most of these systems focus only on tracking usage without providing detailed analysis or

insights. They may show how much time is spent on a particular activity but do not evaluate whether that time is productive or distracting. This limits their effectiveness in improving user behavior.

In comparison, the Digital Attention Span Analyzer provides a more comprehensive approach by analyzing both productive and distraction time. It goes beyond simple tracking by calculating an attention score and categorizing performance levels. This gives users a clear understanding of their focus and helps them identify areas that need improvement. Additionally, the system includes features such as routine analysis and motivation, which are not commonly found in basic tracking tools.

Another key difference is the use of visualization and user-friendly design. While some existing systems may provide data in raw format, this project presents information using charts and graphical representations. This makes it easier for users to interpret their performance and track progress over time. Overall, the proposed model offers a more complete and effective solution compared to traditional and existing systems.

Feature	Content-Based Filtering	Collaborative Filtering	AI Models
Accuracy	Low	Medium	High
Speed	Fast	Fast	Slow
Complexity	Low	Medium	High
Data Need	No data	Small data	Large data
Implementat ion	Easy	Moderate	Difficult

4.3 Advantages and Disadvantages

The Digital Attention Span Analyzer offers several advantages that make it a useful and practical system for improving focus and productivity. One of the main advantages is its **user-friendly interface**, which allows users to easily interact with the system without requiring technical knowledge. The system provides **real-time feedback** in the form of attention scores and performance levels, helping users understand their focus instantly. It also includes **visualization features** such as charts and graphs, which make it easier to analyze performance over time. Another important advantage is its **flexibility**, as it can be used by students, professionals, and anyone interested in improving their concentration. The system is lightweight, cost-effective, and does not require advanced hardware or software, making it highly accessible. Additionally, it is designed with **future scalability**, allowing integration of machine learning models for more accurate predictions and personalized recommendations.

However, the system also has certain limitations that need to be considered. The current implementation relies on **simulated data** rather than real-time user behavior, which may not always reflect actual attention levels accurately. It does not track real-world factors such as mobile usage, notifications, or environmental distractions, which can affect user focus. Another limitation is that the system uses a **rule-based approach**, which lacks the intelligence and adaptability of advanced AI models. This may reduce its ability to handle complex user behavior patterns. Additionally, the system depends on user input for routine analysis, which may sometimes be inaccurate or inconsistent. These disadvantages can be addressed in future enhancements by integrating real-time data collection, sensor-based tracking, and machine learning techniques to improve accuracy and overall system performance.

4.4 Summary of Relevance

The Digital Attention Span Analyzer is highly relevant in today's fast-paced digital world, where maintaining focus has become a major challenge due to the continuous presence of distractions such as smartphones, social media, online entertainment, and multitasking environments. The project directly addresses this real-world issue by providing a systematic and technology-driven approach to monitor and evaluate user attention levels. By tracking productive and distraction time and converting it into a meaningful attention score, the system helps users clearly understand their behavior and identify areas where improvement is required. Unlike traditional methods that rely only on self-discipline, this system introduces a structured mechanism for analyzing attention, making it more effective and reliable. The inclusion of features such as routine analysis, motivational support, and data visualization further enhances its usefulness by guiding users towards better habits and consistent performance.

The relevance of the project is also evident in its wide applicability across different domains such as education, professional work environments, and personal productivity management. Students can use the system to improve their concentration during study sessions, while professionals can utilize it to enhance efficiency and reduce time wastage during work. Moreover, the project demonstrates the practical implementation of artificial intelligence concepts in solving everyday problems, making it academically and technically significant. It also provides a strong foundation for future advancements, where real-time data collection, behavior analysis, and machine learning models can be integrated to create a more intelligent and adaptive system. As digital distractions continue to grow, systems like this will play an important role in promoting focus, discipline, and productivity. Therefore, the Digital Attention Span Analyzer is not only relevant for the present but also holds great potential for future development and real-world impact.

CHAPTER 5

MODULE DESCRIPTION

5.1 User Authentication Module

The User Authentication Module acts as the initial entry point of the Digital Attention Span Analyzer system. It allows users to log in by entering their name, which is then stored using session state. This ensures that each user session is uniquely identified and managed throughout the interaction with the system. Although the authentication process is simple, it plays a crucial role in maintaining user-specific data and enabling personalized tracking of attention levels. It also ensures that the system can maintain continuity of data during the session without requiring repeated inputs.

This module is essential for managing user flow within the application. Once the user successfully logs in, they are redirected to the main dashboard where all other modules are accessible. The session management mechanism ensures that user data such as tracking history and inputs are preserved during usage. In future enhancements, this module can be expanded to include secure authentication methods such as passwords or database integration, making it more robust and suitable for real-world applications.

5.2 Routine Analysis Module

The Routine Analysis Module is designed to evaluate the daily habits of users and understand how these habits influence their attention span. It collects inputs such as sleep duration, working hours, and major distractions like phone usage or social media. Based on these inputs, the system analyzes the routine and provides suggestions to improve productivity and maintain a healthier balance between work and rest. This helps users become more aware of their lifestyle and its impact on their focus.

This module plays a significant role in guiding users toward better habits by highlighting areas that need improvement. For example, if the user has insufficient sleep or low work hours, the system provides warnings and suggestions accordingly. By offering simple yet effective feedback, the module encourages users to adopt disciplined routines. Over time, this contributes to improved attention span and overall efficiency, making it an important supporting component of the system.

5.3 Motivation Module

The Motivation Module is designed to keep users inspired and mentally engaged while using the system. It provides motivational quotes and messages that encourage users to stay focused, consistent, and disciplined in their tasks. These messages are randomly displayed, ensuring variety and keeping the user experience fresh and engaging. This module adds a positive psychological element to the system, which is essential for maintaining long-term productivity.

In addition to boosting motivation, this module helps users overcome discouragement, especially when their attention score is low. Instead of feeling demotivated, users are encouraged to improve and continue their efforts. This promotes a growth mindset and reinforces the idea that consistency leads to success. By combining technical analysis with motivational support, the system ensures a balanced approach to improving attention.

5.4 Attention Tracking Module

The Attention Tracking Module is the core and most important component of the Digital Attention Span Analyzer. It is responsible for monitoring user focus during a selected time duration. The user initiates a tracking session by selecting a time interval, after which the system simulates user behavior by categorizing actions into productive and distraction activities. This simulation helps in estimating how effectively the user is utilizing their time.

At the end of the session, the system calculates an attention score based on the ratio of productive time to total time. This score is then classified into levels such as High, Medium, or Low, providing a clear understanding of the user's performance. The module also records session data, which is stored for further analysis. This real-time feedback mechanism makes the module highly effective in helping users identify their focus levels and take corrective actions immediately.

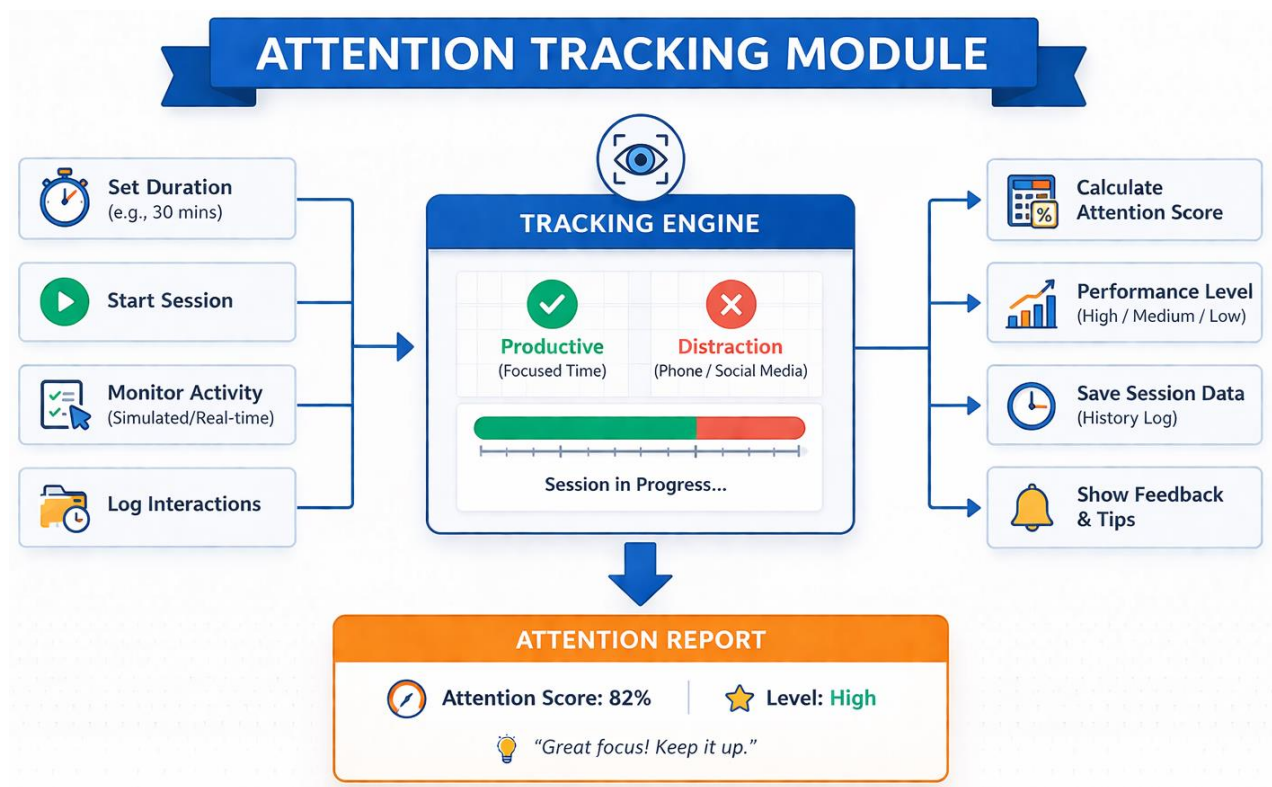


Fig 5.4.1 Attention Tracking Module

5.5 Analytics Module

The Analytics Module is responsible for presenting user performance data in a visual and easy-to-understand format. It uses the data collected from multiple tracking sessions to generate charts such as bar graphs and area charts. These visualizations help users analyze their productivity patterns and understand how their attention levels change over time.

This module plays a crucial role in decision-making by providing insights into user behavior. By observing trends and patterns, users can identify whether their focus is improving or declining. The graphical representation simplifies complex data and makes it more accessible to users. This helps in setting goals and tracking progress effectively, thereby enhancing the overall usefulness.

5.6 Resource Recommendation Module

The Resource Recommendation Module provides users with valuable learning materials that can help improve their focus and productivity. It suggests books and resources such as “Deep Work” and “Atomic Habits,” which are widely recognized for their effectiveness in building discipline and concentration. These resources guide users in developing better habits and strategies for maintaining attention.

This module enhances the system by going beyond analysis and offering solutions for improvement. It encourages users to explore proven techniques and apply them in their daily lives. By integrating educational content with the application, the system provides a holistic approach to attention improvement, combining both technological and behavioral aspects.

5.7 Streamlit Application (central controller)

The Streamlit Application acts as the central controller that integrates all modules into a single unified system. It manages the overall workflow of the application, including user navigation, input handling, and output display. The sidebar menu allows users to easily switch between different modules, ensuring a smooth and interactive experience. This module ensures proper communication between all components of the system. It controls how data flows from one module to another and ensures that all functionalities work together efficiently. The use of Streamlit makes the application dynamic and responsive, allowing real-time updates and seamless interaction. As the backbone of the system, it plays a crucial role in delivering a user-friendly and efficient attention analysis platform.

CHAPTER 6

RESULT AND DISCUSSION

6.1 Performance Analysis

The performance of the Digital Attention Span Analyzer is evaluated based on its ability to effectively monitor user attention, calculate accurate scores, and present meaningful results. During the execution of the system, the Attention Tracking module successfully simulated user behavior by categorizing time into productive and distraction activities. The attention score generated at the end of each session reflected the user's focus level in a clear and understandable manner. The classification of results into performance levels such as High, Medium, and Low further simplified the interpretation of outcomes, making it easier for users to assess their productivity.

The system also demonstrated strong performance in terms of responsiveness and user interaction. The use of progress bars and real-time updates during the tracking session created an engaging experience for users. The application operated smoothly without any significant delays or errors, as it is designed to be lightweight and efficient. The integration of different modules ensured that data flowed seamlessly from tracking to analysis, allowing users to receive immediate feedback after each session.

In addition, the Analytics module enhanced the overall performance evaluation by providing graphical representations of user data. Charts such as bar graphs and area charts helped in identifying patterns and trends over multiple sessions. These visualizations made it easier to understand improvements or declines in attention levels over time. Although the system currently uses simulated data, the results indicate that the model is effective in analyzing attention patterns and can be further improved by incorporating real-time data and advanced algorithms.

6.2 User Feedback

User feedback plays a vital role in understanding the effectiveness and usability of the Digital Attention Span Analyzer. Based on initial testing, users found the system to be simple, user-friendly, and easy to navigate. The interface developed using Streamlit provided a smooth and interactive experience, allowing users to access different modules without confusion. Features such as attention score display, performance levels, and visual charts were well-received, as they helped users quickly understand their focus levels and progress.

Many users appreciated the inclusion of motivational messages, as they added a positive and encouraging aspect to the system. These messages helped users stay motivated even when their performance was low, promoting a mindset of continuous improvement. The Routine Analysis module was also considered useful, as it provided suggestions based on user habits, helping users identify areas where they needed to improve their daily routines.

However, users also suggested certain improvements to enhance the system further. Some users expressed the need for real-time tracking of actual activities instead of simulated data, which would provide more accurate results. Others recommended adding features such as reminders, notifications, and personalized suggestions based on user behavior. Despite these suggestions, the overall feedback was highly positive, indicating that the system is effective in helping users improve their attention span and productivity.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Summary of Outcomes

The Digital Attention Span Analyzer project successfully achieved its primary objective of designing and developing a system that can monitor and evaluate user attention in a digital environment. The system was able to simulate user activity, differentiate between productive and distraction time, and generate an attention score that reflects the user's level of focus. The classification of performance into levels such as High, Medium, and Low made it easier for users to interpret their results. This outcome demonstrates that the system is capable of providing a clear and structured analysis of attention using simple yet effective techniques.

Another important outcome of the project is the successful integration of multiple modules into a single interactive platform. Modules such as user authentication, routine analysis, motivation, attention tracking, analytics, and resource recommendation worked together seamlessly to provide a complete user experience. The use of Streamlit enabled the development of a responsive and user-friendly interface, allowing smooth navigation and real-time interaction. The Analytics module, in particular, helped visualize user performance through charts, making it easier to identify patterns and track progress over time.

In addition, the project highlights the practical application of programming and basic artificial intelligence concepts in solving real-world problems. It demonstrates how data analysis, visualization, and rule-based logic can be combined to create a functional and useful system. Although the current implementation uses simulated data, the results indicate that the system has strong potential for further development. Overall, the project outcomes confirm that the Digital Attention Span Analyzer is an effective tool for improving focus, productivity, and time management in a digital environment.

7.2 Future Scope and Enhancements

The Digital Attention Span Analyzer has significant potential for future improvements by integrating advanced technologies and real-time monitoring capabilities. One of the major enhancements would be the implementation of real-time data collection using system-level tracking, mobile usage analysis, or browser activity monitoring. This would allow the system to capture actual user behavior instead of relying on simulated data, thereby improving the accuracy and reliability of attention analysis. Additionally, integrating sensors or wearable devices can help monitor physical indicators such as eye movement or screen engagement, providing deeper insights into user attention patterns.

Another important area for enhancement is the incorporation of machine learning and artificial intelligence techniques. By using algorithms such as classification and predictive modeling, the system can learn from user behavior and provide personalized recommendations. For example, it can suggest optimal study times, identify peak productivity hours, and detect patterns of distraction. The system can also be enhanced with adaptive learning capabilities, where it continuously improves its predictions based on user data. Furthermore, integrating a database system would enable long-term data storage and advanced analytics, allowing users to track their progress over weeks or months.

In addition to technical improvements, the system can be enhanced by adding more user-centric features to improve engagement and usability. Features such as notifications, reminders, goal setting, and daily progress reports can help users stay consistent and motivated. The development of a mobile application would increase accessibility and allow users to monitor their attention anytime and anywhere. Advanced visualization techniques such as dashboards and detailed reports can provide deeper insights into performance. Future versions can also include multi-user support with secure authentication, making the system suitable for institutions or organizations. With these enhancements, the Digital Attention Span Analyzer can evolve into a powerful and intelligent tool for productivity and attention management.

APPENDICES

APPENDIX A

SOURCE CODE

```
import streamlit as st
import pandas as pd
import time
import random

# ----- CONFIG -----

st.set_page_config(page_title="Attention AI", layout="wide")

# ----- STYLE -----

st.markdown("""
<style>
body {
    background: linear-gradient(135deg,#0f172a,#1e293b);
    color: white;
}
.stButton>button {
    background-color: #22c55e;
    color: white;
    border-radius: 10px;
}
</style>
""", unsafe_allow_html=True)

# ----- SESSION -----

if "user" not in st.session_state:
    st.session_state.user = None
```

```

if "history" not in st.session_state:
    st.session_state.history = []

# ----- LOGIN -----
if st.session_state.user is None:
    st.title("🧠 Attention AI")

    name = st.text_input("Enter your name")

    if st.button("Login"):
        if name.strip() != "":
            st.session_state.user = name
            st.rerun()
        else:
            st.warning("Please enter name")

# ----- MAIN APP -----
else:
    user = st.session_state.user

    st.sidebar.title(f"👤 {user}")

    page = st.sidebar.radio("Menu", [
        "🏠 Home",
        "📝 Routine",
        "💬 Motivation",
        "▶ Tracking",
        "📊 Analytics",
        "📖 Resources"
    ])

```

```

# ----- HOME -----
if page == "🏠 Home":
    st.title(f"🔗 Welcome {user}")

    st.write("Your Focus Journey Starts Now 🧠")

    progress = st.progress(0)
    for i in range(100):
        time.sleep(0.005)
        progress.progress(i + 1)

    st.success("Stay consistent. You will win 🏆")

# ----- ROUTINE -----
elif page == "📅 Routine":
    st.title("📅 Routine Analysis")

    sleep = st.slider("Sleep Hours", 0, 10, 6)
    work = st.slider("Work Hours", 0, 12, 4)
    distraction = st.selectbox("Main Distraction",
                              ["Phone", "Social Media", "Overthinking"])

    if st.button("Analyze Routine"):
        if sleep < 6:
            st.warning("Sleep more 😴")

        if work < 2:
            st.warning("Increase study time 📖")

        if distraction == "Social Media":
            st.warning("Reduce social media 📵")

```

```

else:
    st.success("Good routine 👍")

# ----- MOTIVATION -----
elif page == "💬 Motivation":
    st.title("💬 Motivation Zone")

    quotes = [
        "Focus on progress, not perfection.",
        "Discipline creates success.",
        "Consistency beats talent."
    ]

    st.info(random.choice(quotes))

    if st.button("I'm Motivated"):
        st.success("🏃 Keep going!")

# ----- TRACKING -----
elif page == "▶ Tracking":
    st.title("▶ Attention Tracking")

    duration = st.slider("Duration (seconds)", 5, 30, 10)

    start = st.button("Start Tracking")

    if start:
        p = 0
        d = 0
        log = []

```



```

progress_bar = st.progress(0)

for i in range(duration):
    if random.random() > 0.5:
        p += 1
    else:
        d += 1

    log.append({"time": i, "productive": p, "distraction": d})

    progress_bar.progress((i + 1) / duration)
    time.sleep(1)

idle = duration - (p + d)

score = int((p / (p + d + 1)) * 100)

if score > 70:
    level = "HIGH 🟢"
elif score > 40:
    level = "MEDIUM 🟡"
else:
    level = "LOW 🔴"

st.subheader("📊 Result")
st.metric("Attention Score", f"{score}/100")
st.success(f"Level: {level}")

df = pd.DataFrame(log)

```

```

st.area_chart(df.set_index("time"))

st.session_state.history.append({
    "productive": p,
    "distraction": d,
    "idle": idle
})

# ----- ANALYTICS -----
elif page == "📊 Analytics":
    st.title("📊 Performance Analytics")

    if st.session_state.history:
        df = pd.DataFrame(st.session_state.history)
        st.bar_chart(df)
    else:
        st.info("No data available")

# ----- RESOURCES -----
elif page == "📖 Resources":
    st.title("📖 Improve Your Focus")

    st.markdown("### 📖 Deep Work")
    st.markdown("[Open Book](https://www.calnewport.com/books/deep-work/)")

    st.markdown("### 📖 Atomic Habits")
    st.markdown("[Open Book](https://jamesclear.com/atomic-habits)")

```

APPENDIX B – Screenshots

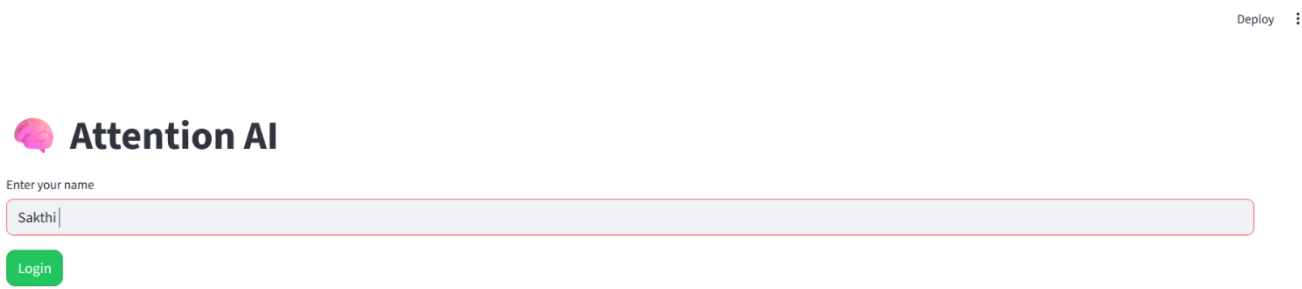


Figure B.1 Login Interface

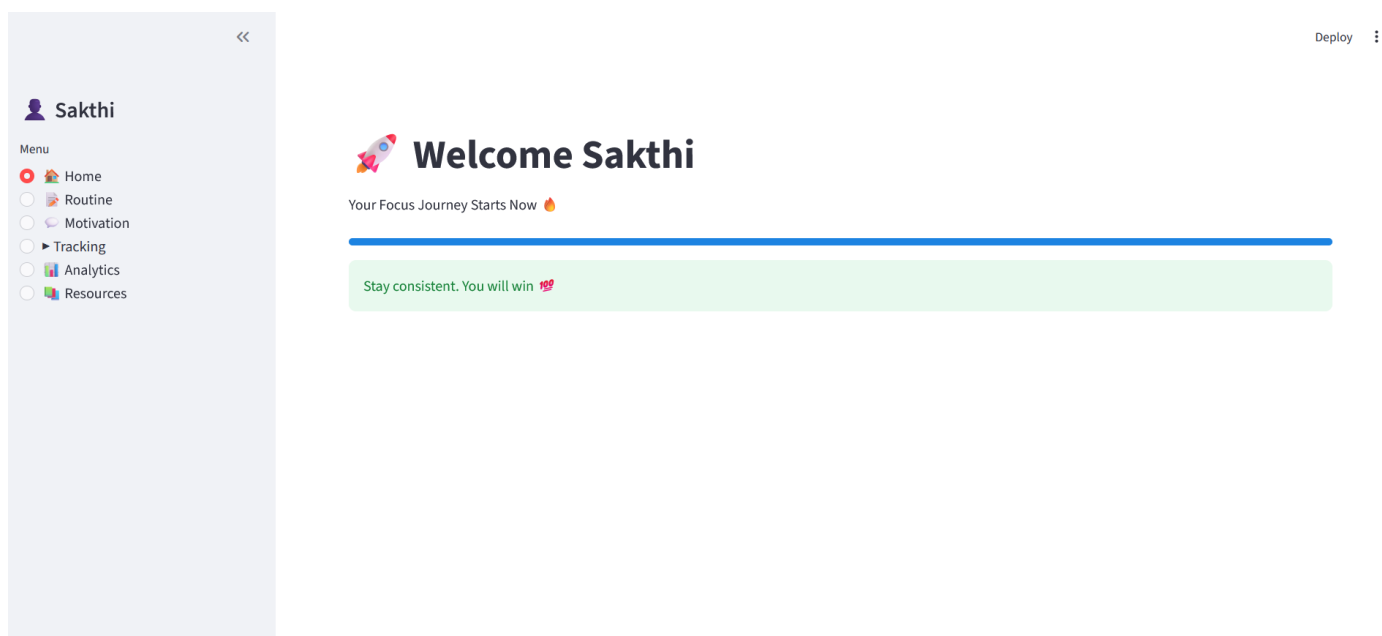


Fig B.2. Home Dashboard Screen

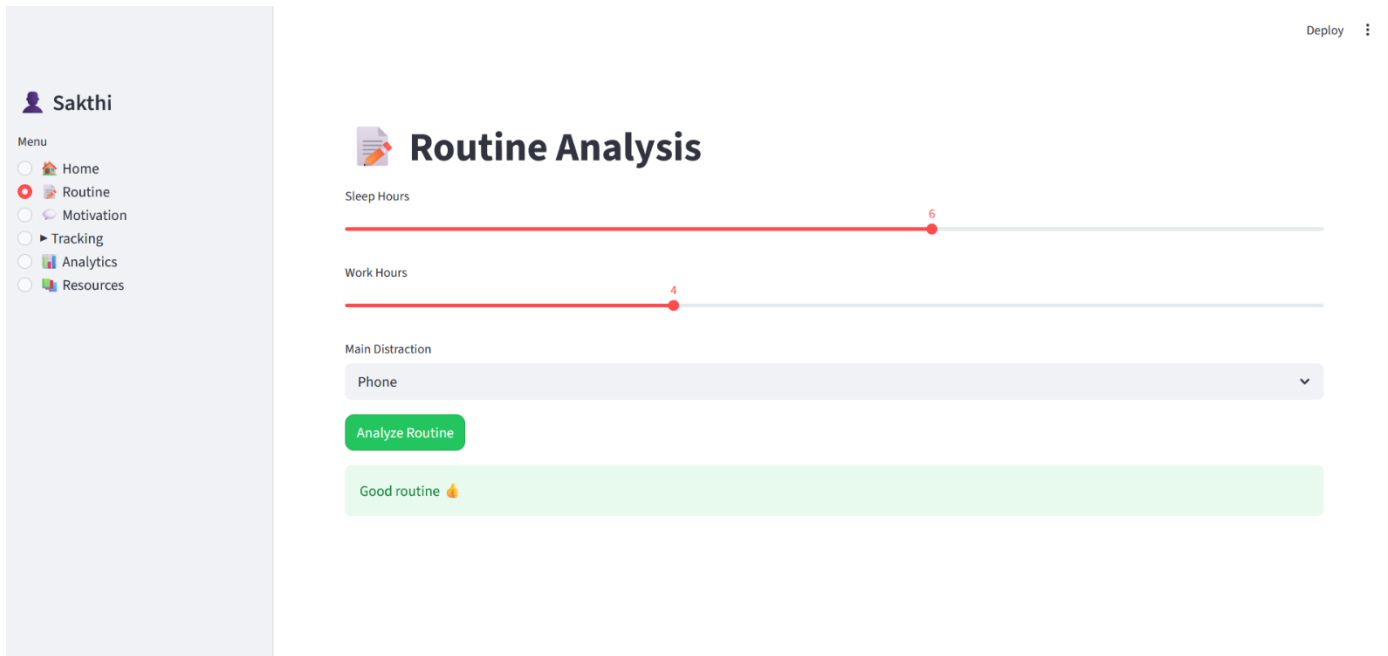


Figure B.3. Routine Analysis Output



Figure B.4. Attention Level Result Screen

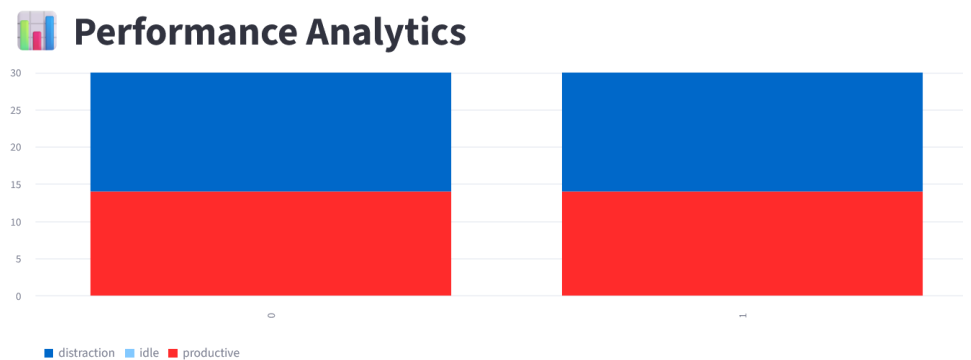
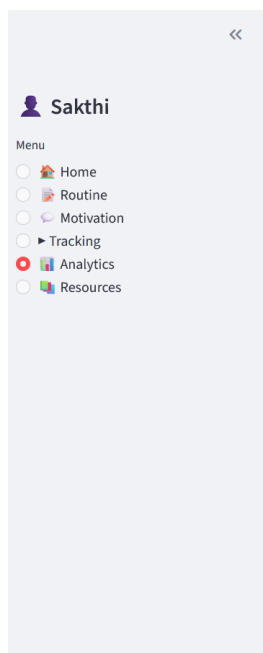


Figure B.5. Attention Score Display

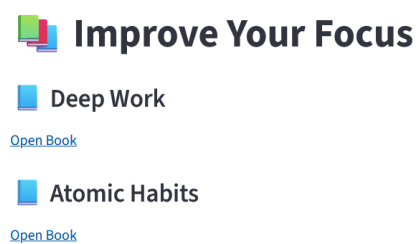
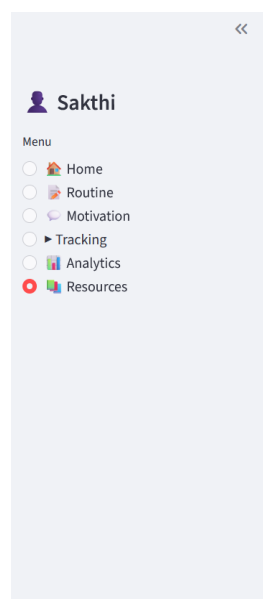


Figure B.6. Resources / Books Page

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